Supplemental Material: Energy-selective four-time correlations in dissipative Majorana systems

OpenQuantum
(Dated: November 8, 2026)

I. CONVENTIONS AND FERMIONIC REGRESSION

Let $N = 2m$, $d = 2^m$, and $\tau(X) = d^{-1}\,\mathrm{tr}\,X$. Majoranas satisfy $\{\psi_i, \psi_j\} = \delta_{ij}$, and the normalized Hilbert–Schmidt inner product is $(A, B) = \tau(A^\dagger B)$. The Hamiltonian $H = \sum_{|A|=q} J_A \psi_A$ has real coefficients, $q \equiv 0 \pmod 4$, and $H \neq 0$. The notation ψ_A means the product in increasing index order. In this convention H is Hermitian and traceless. These assumptions apply to the homogeneous model; the mixed-order example in Sec. VI uses Hermitian interaction components of different even degrees.

For $L_i = \sqrt{\mu}\,\psi_i$, the physical Lindblad dynamics [1, 2] gives the Heisenberg generator on even operators, while fermionic regression specifies the graded generator on odd operators:

$$\mathcal{L}_e X = i[H, X] - \mu \hat{n} X, \qquad \hat{n} X = \frac{N}{2} X - \sum_i \psi_i X \psi_i, \tag{S1}$$

$$\mathcal{L}_o X = i[H, X] - \mu \left(\frac{N}{2} X + \sum_i \psi_i X \psi_i \right). \tag{S2}$$

Their Hilbert–Schmidt adjoints are denoted $\mathcal{M}_e = \mathcal{L}_e^\dagger$ and $\mathcal{M}_o = \mathcal{L}_o^\dagger$; they reverse the sign of $i[H, \cdot]$ and retain the dissipative part. Each generator preserves Hermitian conjugation. The maximally mixed state is stationary. Pairing Majoranas into $a_r = (\psi_{2r-1} + i\psi_{2r})/\sqrt{2}$ shows that the two Majorana dissipators are equivalent to jumps $\sqrt{\mu}\,a_r$ and $\sqrt{\mu}\,a_r^\dagger$.

Equation (S2) is the fermionic-regression extension, not the action of the ungraded spin Lindbladian on odd Jordan–Wigner matrices. An odd insertion requires a minus sign on the recycling term while leaving the anticommutator term unchanged [3, 4]. This gives a single-Majorana dissipative rate μ; the ungraded odd extension would instead give $(N-1)\mu$. The modified odd evolution acts on auxiliary operators in correlation functions, not on physical density matrices. Markovian regression [5] is applied successively in chronological order using its multitime extension [6].

For $t_1 < t_2 < t_3 < t_4$, write $s = t_2 - t_1$, $g = t_3 - t_2$, and $u = t_4 - t_3$. Literal four-insertion regression gives

$$G_4(s, g, u) = \frac{1}{N^2} \sum_{i,j} \tau\left[\psi_i e^{u\mathcal{M}_o}\left\{\psi_i e^{g\mathcal{M}_e}\left[\psi_j e^{s\mathcal{M}_o}(\psi_j)\right]\right\}\right]. \tag{S3}$$

The initial factor I/d is included in τ. Moving the last evolution to the readout defines

$$R_u = \frac{1}{N}\sum_i (e^{u\mathcal{L}_o}\psi_i)\psi_i, \qquad S_s = \frac{1}{N}\sum_j \psi_j (e^{s\mathcal{M}_o}\psi_j), \qquad G_4 = \tau[R_u e^{g\mathcal{M}_e} S_s]. \tag{S4}$$

Thus R and S are fixed by the four insertions. They cannot be replaced by an arbitrary composite observable or by a Hermitian autocorrelation. In particular, $R_s = S_s$ at finite s is not assumed.

Define $C_N(s) = N^{-1}\sum_i \tau[\psi_i (e^{s\mathcal{L}_o}\psi_i)]$. Each summand is real because both factors are Hermitian. Adjointness then gives

$$\tau R_s = \tau S_s = C_N(s), \qquad C_N(0) = \frac{1}{2}. \tag{S5}$$

The pair-connected quantity used throughout is $G_{\mathrm{pair}} = G_4 - C_N(u)C_N(s)$. It removes the stationary projection during g. It is not the fully Wick-subtracted fermion cumulant. For example, in a flavor-diagonal approximation the additional cross-pair subtraction involves

$$\frac{C_N(g+s+u)C_N(g) - C_N(g+u)C_N(g+s)}{N}. \tag{S6}$$

Its mixed derivative at $s = u = 0$ is $[C_N C_N'' - (C_N')^2]/N$, which is generally nonzero. This term must be handled explicitly in any comparison with a different convention for a connected ladder response. Finally, the source of an ordinary Hamiltonian Kubo response [7] at infinite temperature is $[V, I/d] = 0$; Eq. (S3) describes a nonzero multitime correlation, not that susceptibility.

II. EXACT ENERGY PROJECTION AND THE SOURCED AMPLITUDE

An even Majorana string of length k is an eigenoperator of the size superoperator $\hat{n}$ with eigenvalue k [8, 9]. Since H is homogeneous and commutes with itself,

$$\mathcal{L}_e H = \mathcal{M}_e H = -q\mu H, \qquad \mathcal{L}_e I = \mathcal{M}_e I = 0. \tag{S7}$$

The projection below uses the exact energy mode established in dissipative SYK work [10] and its general common-eigenoperator mechanism [11].

Set $v = \tau H^2 > 0$ and define the orthogonal projections

$$P_I X = \tau(X) I, \qquad P_H X = H \frac{\tau(HX)}{v}, \qquad Q = 1 - P_I - P_H. \tag{S8}$$

Equation (S7) holds for both generators, so P_I, P_H, Q commute with $\mathcal{L}_e$ and $\mathcal{M}_e$. No spectral simplicity, diagonalizability of the complementary block, or energy-gap ordering is required. Applying these projections to Eq. (S4) gives

$$G_4(s,g,u) = C_N(u)C_N(s) + \frac{b_R(u)b_S(s)}{v} e^{-q\mu g} + G_Q(s,g,u), \tag{S9}$$

$$b_R(u) = \tau(HR_u), \qquad b_S(s) = \tau(HS_s), \qquad G_Q = \tau[(QR_u)e^{g\mathcal{M}_Q}(QS_s)], \qquad \mathcal{M}_Q = Q\mathcal{M}_e Q. \tag{S10}$$

The notation G_Q here is $G_\perp$ in the main text. Its separation from the energy direction is exact even when another mode has eigenvalue $-q\mu$. Such a mode stays in G_Q. The energy contribution to the gap-Laplace transform is

$$\widehat{G}_H(z;s,u) = \frac{b_R(u)b_S(s)}{v(z+q\mu)}. \tag{S11}$$

The residue is a product, not $|b|^2$. It need not be positive or real. It is also not necessarily the total residue of the pole when the complementary block is degenerate with energy.

III. ANTIUNITARY SYMMETRY AND THE INFORMATION IN A TWO-POINT FUNCTION

The scalar source and readout vertices are equal under the stated full-space trace convention. Choose Jordan–Wigner matrices $\gamma_i = \sqrt{2}\psi_i$ with γ_{2r-1} real and γ_{2r} imaginary. Let K be complex conjugation and

$$\Theta = UK, \qquad U = \gamma_2\gamma_4\cdots\gamma_{2m}, \qquad \Theta\psi_i\Theta^{-1} = (-1)^m\psi_i. \tag{S12}$$

Conjugation by U gives sign $(-1)^m$ outside this product and $(-1)^{m-1}$ inside it; K supplies the additional minus sign in the latter case. Real couplings and $q \equiv 0 \pmod 4$ imply $\Theta H\Theta^{-1} = H$. The induced antilinear map $\mathcal{T}(X) = \Theta X\Theta^{-1}$ therefore satisfies $\mathcal{T}\mathcal{L}_o = \mathcal{M}_o\mathcal{T}$. Since the generators preserve Hermitian conjugation,

$$\mathcal{T}(R_s) = \frac{1}{N}\sum_i (e^{s\mathcal{M}_o}\psi_i)\psi_i = S_s^\dagger. \tag{S13}$$

Using $\tau(\mathcal{T}X) = \tau(X)^*$ and trace cyclicity,

$$b_R(s)^* = \tau[\mathcal{T}(HR_s)] = \tau(HS_s^\dagger) = b_S(s)^*, \qquad b_R(s) = b_S(s) \equiv b_N(s). \tag{S14}$$

This argument uses the full Hilbert-space trace and imposes no restriction on N mod 8. It does not imply $R_s = S_s$. With $P = i^m\gamma_1\cdots\gamma_N$, one has $\Theta P\Theta^{-1} = (-1)^m P$; thus a single-parity-block trace requires care when Θ exchanges the blocks. We do not use a block-restricted correlation.

Write $X_i(s) = e^{s\mathcal{L}_o}\psi_i$. Self-adjointness of the dissipative part and $\mathcal{D}_o\psi_i = -\mu\psi_i$ give

$$\begin{aligned} C_N'(s) + \mu C_N(s) &= \frac{i}{N}\sum_i \tau\{H[X_i(s)\psi_i - \psi_i X_i(s)]\} \\ &= i[b_N(s) - b_N(s)^*]. \end{aligned} \tag{S15}$$

Consequently, for $c_N = 2C_N$,

$$b_N(s) = x_N(s) + \frac{c'_N(s) + \mu c_N(s)}{4i}, \qquad x_N(s) = \mathrm{Re}\, b_N(s). \tag{S16}$$

Only the imaginary vertex is fixed by C_N. The second-order expansion derived in Sec. IV gives

$$b_N(s) = \frac{iqv}{N}s - \frac{q\tau(H^3)}{2N}s^2 - \frac{i\mu q^2 v}{2N}s^2 + O(s^3). \tag{S17}$$

The real term need not vanish. For example, with $Z_r = 2i\psi_{2r-1}\psi_{2r}$, the pure quartic Hamiltonian $H = aZ_1Z_2 + bZ_2Z_3 + cZ_1Z_3$ has $\tau H^3 = 6abc$.

For an ensemble invariant under $H \mapsto -H$, reversing H interchanges $\mathcal{L}_o$ and $\mathcal{M}_o$ and gives $b_{-H}(s) = -b_H(s)^*$. Hence $\overline{x_N} = 0$, but with $d_N = c'_N + \mu c_N$,

$$\frac{\overline{b_N(u)b_N(s)}}{v} = \frac{\overline{x_N(u)x_N(s)}}{v} - \frac{1}{16}\frac{\overline{d_N(u)d_N(s)}}{v}. \tag{S18}$$

Symmetry removes the imaginary cross terms, not the real-vertex covariance. A large-N factorization of this expression requires more than symmetry.

IV. SHORT-PAIR EXPANSION AND POINT-SPLIT ENERGY SELECTION

Define $\Psi(X) = \sum_i \psi_i X \psi_i$. For a length-$k$ string X_k,

$$\Psi(X_k) = (-1)^k \left(\frac{N}{2} - k\right) X_k. \tag{S19}$$

On even strings $\Psi = N/2 - \hat{n}$; on odd strings Eq. (S2) has dissipative action $-\mu k$. Let $A_i = [H, \psi_i]$, which has degree $q - 1$. Then

$$\begin{aligned} \mathcal{L}_o^2\psi_i &= -[H,[H,\psi_i]] - i\mu q[H,\psi_i] + \mu^2\psi_i, \\ \mathcal{M}_o^2\psi_i &= -[H,[H,\psi_i]] + i\mu q[H,\psi_i] + \mu^2\psi_i. \end{aligned} \tag{S20}$$

Using $\Psi(H) = (N/2 - q)H$ yields

$$\sum_i [H,\psi_i]\psi_i = qH, \qquad \sum_i \psi_i[H,\psi_i] = -qH. \tag{S21}$$

Thus the pair vertices obey

$$R_0 = S_0 = \frac{I}{2}, \qquad R'_0 = S'_0 = \frac{iq}{N}H - \frac{\mu}{2}I. \tag{S22}$$

The full flavor sum is used in Eq. (S21). A single flavor gives only the interaction terms containing it; omitting diagonal pairs $i = j$ from G_4 also removes the factorized full sum that enables the cancellation.

For the double commutators,

$$\begin{aligned} \sum_i [H,[H,\psi_i]]\psi_i &= \frac{N}{2}H^2 - 2H\Psi(H) + \Psi(H^2) = (2q - \hat{n})H^2, \\ \sum_i \psi_i[H,[H,\psi_i]] &= \Psi(H^2) - 2\Psi(H)H + \frac{N}{2}H^2 = (2q - \hat{n})H^2. \end{aligned} \tag{S23}$$

Together with Eq. (S20), this proves

$$R''_0 = S''_0 = \frac{(\hat{n} - 2q)H^2 - i\mu q^2 H}{N} + \frac{\mu^2}{2}I. \tag{S24}$$

Tracing this identity against H and using self-adjointness of $\hat{n}$ gives Eq. (S17). Projecting instead with Q defines

$$B = Q[(\hat{n} - 2q)H^2],$$
$$QR_s = \frac{s^2}{2N}B + O(s^3), \qquad QS_s = \frac{s^2}{2N}B + O(s^3). \tag{S25}$$

These are equal coefficients through second order, not equality of the full finite-time operators. Equivalently,

$$B = (\hat{n} - 2q)H^2 + 2qvI + q\frac{\tau(H^3)}{v}H. \tag{S26}$$

For $q = 4$, symmetrized products of two quartic strings vanish for odd overlap, so H^2 contains only lengths $0, 4, 8$. If Π_k projects onto length k,

$$H^2 = vI + \Pi_4(H^2) + \Pi_8(H^2), \qquad B = -4\left[\Pi_4(H^2) - \frac{\tau(H^3)}{v}H\right]. \tag{S27}$$

The length-eight term is absent when $N < 8$.

Equation (S22) also isolates energy exactly in a point-split derivative. Applying $(\partial_s + \mu)$ and $(\partial_u + \mu)$ to the separated vertices first gives

$$(\partial_u + \mu)(\partial_s + \mu)G_4(s, g, u)|_{s=u=0^+} = -\frac{q^2 v}{N^2}e^{-q\mu g}. \tag{S28}$$

The stationary and Q components vanish. This is the known energy autocorrelation expressed through the short-pair insertions; the finite-duration error is the additional issue addressed next. The one-sided limits are taken after specifying strictly separated chronological times. Differentiating a globally time-ordered distribution also produces contact terms from fermion anticommutators, which are not included in Eq. (S28).

V. COMPLEMENTARY CONTRIBUTION AND FINITE-TIME BOUND

Both complementary vertices start at second order, which fixes the order of their contribution to the correlation. Substituting Eq. (S25) into Eq. (S10) gives, at fixed finite N, H, μ, g,

$$G_Q(s, g, u) = \frac{s^2u^2}{4N^2}\tau[Be^{g\mathcal{M}_Q}B] + O(s^3u^2 + s^2u^3). \tag{S29}$$

Meanwhile the energy amplitude starts at $-q^2vsu/N^2$. For equal pairs, $G_Q = O(\epsilon^4)$ and the energy amplitude is $O(\epsilon^2)$, proving the relative $O(\epsilon^2)$ selection rule. To isolate G_Q at finite duration, one must subtract the complete amplitude $b_R(u)b_S(s)/v$, not only its leading term: the energy vertices themselves have cubic corrections when $s = u$.

A bound valid at every nonnegative g follows without resolving the complementary spectrum. Every traceless even string has length at least two. For X in the range of Q,

$$\mathrm{Re}(X, \mathcal{M}_Q X) = -\mu(X, \hat{n}X) \le -2\mu\|X\|_2^2. \tag{S30}$$

Integrating the differential inequality for the squared norm yields

$$\|e^{g\mathcal{M}_Q}X\|_2 \le e^{-2\mu g}\|X\|_2, \qquad |G_Q| \le e^{-2\mu g}\|QR_u\|_2\|QS_s\|_2. \tag{S31}$$

In the last step one uses $\tau(AY) = (A^\dagger, Y)$ and $\|A^\dagger\|_2 = \|A\|_2$. This estimate uses the Hermitian part of the generator, as in spectral bounds [12] and transient-correlation bounds [13]. It remains valid for a nonnormal complementary block.

When $b_R(u)b_S(s) \neq 0$, division by the exact energy contribution gives

$$\frac{|G_Q|}{|G_H|} \le \frac{v\|QR_u\|_2\|QS_s\|_2}{|b_R(u)b_S(s)|}e^{(q-2)\mu g}. \tag{S32}$$

For equal sufficiently short pairs this becomes

$$\frac{|G_Q|}{|G_H|} \le \left[\frac{\epsilon^2\|B\|_2^2}{4q^2v} + O(\epsilon^3)\right]e^{(q-2)\mu g}. \tag{S33}$$

The figures evaluate the exact bound in Eq. (S32); the truncated expansion explains its short-pair scaling. Finite-dimensional analyticity bounds the vertex Taylor remainders on a short interval. Combined with Eq. (S31), it gives the absolute remainder in Eq. (S29) as $O(e^{-2\mu g}(s^3u^2+s^2u^3))$, with a constant independent of $g \geq 0$ for the fixed system. Uniformity of this constant in N would require a separate estimate. Because energy decays faster than the guaranteed complementary contraction rate, the relative bound can grow with g: it controls finite-delay selection and does not establish asymptotic energy dominance.

Orthogonality of distinct Majorana strings gives the exact sample identity $v = 2^{-q}\sum_A J_A^2$. If $\overline{J_A^2} = J^2/\binom{N-1}{q-1}$,

$$\overline{v} = \frac{NJ^2}{q2^q}, \qquad \overline{G_H(\epsilon,g,\epsilon)} = -\frac{qJ^2}{N2^q}\epsilon^2 e^{-q\mu g} + O(\epsilon^3). \tag{S34}$$

For the finite-N Gaussian ensemble used here the necessary moments are finite. The variance formula itself requires neither independence nor Gaussianity. Equation (S34) gives the averaged short-time coefficient, not a factorization of the finite-duration residue.

VI. MIXED INTERACTION ORDERS

Let $H = H_q + H_p$, where H_q, H_p are traceless Hermitian Majorana interactions of distinct even lengths. The first-order identity extends to

$$(\partial_s+\mu)R_s|_{0^+} = (\partial_s+\mu)S_s|_{0^+} = \frac{i}{N}\hat{n}H = \frac{i}{N}(qH_q+pH_p). \tag{S35}$$

Here R, S are generated with the full mixed Hamiltonian. The total energy obeys $\mathcal{L}_e H = -\mu(qH_q+pH_p)$ and generally is not an eigenoperator. Thus the pure-q pole formula cannot be applied to total H.

Write $v_q = \tau H_q^2$, $v_p = \tau H_p^2$ and $Q_H X = X - H\tau(HX)/(v_q+v_p)$ on the traceless space. String orthogonality gives

$$\begin{aligned}\|Q_H\hat{n}H\|_2^2 &= q^2v_q + p^2v_p - \frac{(qv_q+pv_p)^2}{v_q+v_p}\\ &= \frac{(q-p)^2v_qv_p}{v_q+v_p}.\end{aligned} \tag{S36}$$

This is a statement about the initial vertex direction, not a spectral decomposition of the mixed model. The control uses $N=8$, $\mu/J=0.2$, and $H = H_4 + 0.25J\,i\psi_0\psi_1$. With $J=1$, $v_4 = 0.14327541260849022$, $v_2 = 0.015625$, the directly evaluated squared norm is 0.05635424817992054, compared with 0.05635424817992057 from Eq. (S36). Its fraction of $\|\hat{n}H\|_2^2$ is 0.02393056613726456. No single-rate total-energy decay is assumed in this control.

VII. RELATION TO THE CONTINUOUS SINGLET LADDER

A large-N singlet-ladder identity provides a separate check of the energy-vertex shape. The finite-N selection theorem follows from the projections and bounds above. In the normalization of the main text, let $C(0) = 1/2$, $c = 2C$, and $a = 2^{2-q}J^2$. The dissipative SYK Schwinger–Dyson equation is [14, 15]

$$c'(t) = -\mu c(t) - a\int_0^t c(t-s)^{q-1}c(s)\,ds. \tag{S37}$$

The doubled-contour components are $G_{LL} = G_{RR} = \mathrm{sgn}(t)C(|t|)$ and $G_{LR} = -G_{RL} = iC(|t|)$. With $\eta_L = -i$, $\eta_R = i$, the standard singlet kernel is [16–18]

$$K_{ab,cd} = -(q-1)J^2\eta_c\eta_d G_{ac}(t_{13})G_{bd}(t_{24})G_{cd}(t_{34})^{q-2}. \tag{S38}$$

For $q \equiv 0 \pmod 4$, all four rung weights coincide. Writing $G = C[\mathrm{sgn}(t)I - \sigma_y]$ and $\boldsymbol{u} = (1,-i)^T$ gives $G(t)\boldsymbol{u} = 2\theta(t)C(t)\boldsymbol{u}$. Thus $\phi_{ab}(s) = u_a u_b F(s)$ is a closed retarded–retarded channel of an ansatz with center-time dependence $e^{-E(t_1+t_2)/2}$. Singlet exchange requires $F(-s) = -F(s)$.

Let $b = E/2$, let $*$ denote whole-line convolution, and define

$$\begin{aligned} R_\pm(s) &= \theta(\pm s)e^{b|s|}c(|s|), & V_\pm(s) &= \theta(\pm s)e^{b|s|}c(|s|)^{q-1},\\ D_\pm &= \pm\partial_s + \mu - b + aV_\pm *, & w(s) &= (q-1)ac(|s|)^{q-2}. \end{aligned} \tag{S39}$$

The rail functions $R_\pm$ are unrelated to the finite-system readout R_s in Eq. (S4). Equation (S37) implies $D_+R_+ = D_-R_- = \delta$ and $K_E F = (R_+ * R_-) * [wF]$. Consider

$$F_E(s) = \mathrm{sgn}(s) e^{E|s|/2}[c'(|s|) + \mu c(|s|)] = -a(V_+ * R_+ - V_- * R_-). \tag{S40}$$

Applying D_+D_- cancels the mixed convolutions $V_+ * V_-$ and yields

$$D_+D_-F_E = a\partial_s(V_+ + V_-) - a(\mu - b)(V_+ - V_-). \tag{S41}$$

The sum $V_+ + V_-$ is continuous at zero, so its derivative contains no delta function. Subtracting wF_E leaves

$$(D_+D_- - w)F_E = a(E - q\mu)(V_+ - V_-). \tag{S42}$$

When the inverse convolutions converge, $E = q\mu$ therefore gives $K_{q\mu}F_{q\mu} = F_{q\mu}$. If $c(t)$ has leading exponential envelope $e^{-\Gamma_1 t}$, sufficient absolute rail convergence is $\mathrm{Re}\, E < 2\Gamma_1$. Provided its leading coefficient is nonzero, $F_{q\mu}$ has envelope $e^{-(\Gamma_1 - q\mu/2)|s|}$. Loss of this integrability is a property of the relative-time representation. An odd projection can cancel a leading constant boundary divergence; the rail condition alone is not a determination of the full continuous spectrum.

To compare with the sourced finite-system correlation, the separation between pair centers is

$$T = g + \frac{s+u}{2}, \qquad G_H = e^{-q\mu T}\frac{\widetilde{b}_N(u)\widetilde{b}_N(s)}{v}, \qquad \widetilde{b}_N(s) = e^{q\mu s/2} b_N(s). \tag{S43}$$

The exponential in Eq. (S40) consequently comes from the gap-to-center-time change of variables. The two-point derivative fixes the imaginary part of this vertex, by Eq. (S16). If, at fixed finite pair durations, one additionally establishes

$$\overline{x_N(u)x_N(s)/v} = o(N^{-1}), \qquad \overline{d_N(u)d_N(s)/v} = d(u)d(s)/\overline{v} + o(N^{-1}), \tag{S44}$$

where $d = c' + \mu c$, then Eq. (S18) implies the conditional formula

$$\overline{G_H} = -\frac{q}{4aN} e^{-q\mu T} F_{q\mu}(u) F_{q\mu}(s) + o(N^{-1}). \tag{S45}$$

Those self-averaging estimates are not proved or tested by the present two-realization finite-size data. They are not used for any main-text result. Moreover, chronological pairs require $T \geq (s+u)/2$; taking arbitrarily large s, u at fixed T leaves the physical domain of Eq. (S3). Neither a homogeneous kernel root nor loss of relative-time integrability establishes disappearance of the exact energy contribution. A complete sourced ladder response must also retain its bare term and the appropriate connected convention [18]. No flavor-complete spectrum or thermodynamic gap transition is inferred here.

VIII. NUMERICAL CHECKS, FIGURE DATA, AND REPRODUCIBILITY

The retained data contain $q = 4$, $N = 6, 8, 10$, two random seeds (1701, 1702), and $\mu/J = 0.05, 0.3228079267, 0.4$, giving 18 systems. The middle noise value is a retained sampling point, not an asserted physical transition. Pair durations are $sJ = 0, 0.02, 0.04, 0.08, 0.16, 0.4, 1$. Four-insertion correlations use equal durations $sJ = uJ = 0.04, 0.08, 0.16$ and an unequal-duration pair $(sJ, uJ) = (0.4, 1)$. The evaluated delays are $gJ = 0, 0.7, 3$. The mixed-order control is specified in Sec. VI.

Hamiltonians use Jordan–Wigner Majoranas and the stated coupling variance. An independent sparse Kronecker construction implements $\mathcal{L}_{e,o}$ and their adjoints on vectorized even and odd matrices. Matrix exponentials act through SciPy's `expm_multiply`. Direct full-Hilbert-space matrix action checks both parity sectors and both adjoint directions. Literal four-insertion propagation checks the grouping into R, S. Hamiltonian construction is shared with the retained SYK implementation, so this is not a fully independent implementation of every model component.

For main-text Fig. 1(b), $N = 10$, seed 1701, and $\epsilon J = 0.16$ are fixed. The three symbols per noise strength are $|G_{\mathrm{pair}}(\epsilon, g, \epsilon)|/|G_{\mathrm{pair}}(\epsilon, 0, \epsilon)|$. Smooth curves are the known energy exponentials, not interpolated additional data or fits. Panel (c) uses the maximum actual contamination and maximum exact bound across all 18 systems, separately at each ϵ. These maxima are recorded in Table S2; they are not statistical confidence bounds.

For main-text Fig. 2, $N = 8$, seed 1701 and $\mu/J = 0.3228079267$. The plotted fractional discrepancy is

$$\frac{|\mathcal{A}_H(\epsilon,\epsilon) - \mathcal{A}_H^{(2)}(\epsilon,\epsilon)|}{|\mathcal{A}_H(\epsilon,\epsilon)|}, \qquad \mathcal{A}_H^{(2)} = -\frac{[C_N'(\epsilon) + \mu C_N(\epsilon)]^2}{4v}. \tag{S46}$$

TABLE S1. Retained finite-matrix checks. Errors refer to the 18 pure quartic systems. Taylor slopes diagnose analytic orders; they are not fitted critical exponents.

Check	Result
Independent generator action, maximum relative error	5.58×10^{-16}
Literal four-insertion propagation versus pair vertices, maximum absolute error	1.12×10^{-16}
Stationary–energy–complementary decomposition, maximum absolute error	5.50×10^{-16}
First/second vertex derivative, maximum norm error	2.26×10^{-16}
Point-split Ward correlation, maximum absolute error	2.78×10^{-17}
Finite-duration slope of $\|QR_s\|_2$	1.920–1.989
Finite-duration slope of $\|G_Q\|$ at equal pairs	3.787–3.972
Finite-duration slope of $\|G_Q\|/\|G_H\|$	1.930–1.992
Exact norm bound	Satisfied at every evaluated point
Warnings and assertion failures	None

TABLE S2. Source data for the finite-duration window, at $gJ = 0.7$. Values are fractions, not percentages.

ϵJ	Maximum $\|G_Q\|/\|G_H\|$	Maximum bound from Eq. (S32)
0.04	$6.2416206393 \times 10^{-5}$	$1.0757258094 \times 10^{-4}$
0.08	$2.4836295296 \times 10^{-4}$	$4.1694373943 \times 10^{-4}$
0.16	$9.8305521244 \times 10^{-4}$	$1.5673453771 \times 10^{-3}$

This approximation retains only the imaginary vertices. At $\epsilon J = 1$, the full value is $-0.008388800296598862 - 0.0005169977038992778i$, while $\mathcal{A}_H^{(2)} = -0.008396758326812267$. The complete complex residue, not its real part alone, is used in this comparison and in the subtraction defining G_Q.

The accompanying project contains the following files:

- Validation code: `code/check_sourced_energy_response.py`.
- Retained data: `data/sourced_energy_response_checks_2026-11-02.json`.
- Figure code: `code/figs_pra_energy_selection.py`.

The figure script reads that input and writes the two figures, a compact JSON provenance record, and CSV tables. It does not rerun dynamics. Running the check script with `python3 -I -B` and `--mixed-control` prints a new validation record without overwriting the retained data. Small-matrix products use `einsum(optimize=False)` to avoid observed local BLAS warnings; warnings are collected and asserted absent, rather than suppressed. Historical spectral calculations and failed extrapolations remain in the project, but do not support the finite-time selection claims presented here.

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